Angala Vikas

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.73
Sri Chaitanya Jr kalasala	Class XII (Telangana State Board)	2023	95.90%
Sri Chaitanya High School	Class X (Telangana State Board)	2021	10.00%

RELEVANT COURSES

Computer Science: Data Structures and Applications, Algorithms, Computer Networks, Software Engineering, Database Management Systems, Operating Systems, Theory of Computation, Discrete Mathematics.

Artificial Intelligence: Machine Learning, Fraud Analytics, AI for Finance,Convex Optimisation,Probability & Random Variables.

WORK EXPERIENCE

AI Engineer Intern | Oracle

Tech: Oracle AI Studio, LLM Workflow Orchestration, Fusion REST APIs, MCP-Style Tool Calling

- Architected an **agentic AI workflow** in **Oracle AI Studio** to autonomously detect and resolve project financial exceptions, self-identifying **4 use cases** from an open-ended brief and validating them with automated **schema and workflow checks**.
- Designed **MCP-style tool calling** with schema-driven **Business Object** abstractions over **Oracle Fusion REST APIs**, letting LLM nodes retrieve enterprise data across **6 interconnected workflows (150+ nodes)** via stage-specific **JSON payloads**.
- Built deterministic **data normalization pipelines** for inconsistent API responses (arrays, nested objects, missing fields) and a configurable **policy node layer**. **Progressive context selection** cut **LLM token usage by 60%** across **4 LOOP-driven** stages.
- Implemented **human-in-the-loop guardrails** with **SWITCH-based routing** to reject malformed action payloads before execution, enforcing a recommend-not-act boundary where all Fusion writes required explicit PM confirmation via **REST-backed function calls**.

PROJECTS

A Comparative Study of SDN Load Balancing Strategies in Data Centre Networks

Tech: Python, Ryu Controller, OpenFlow 1.3, Mininet, NetworkX, TCP/IP, iperf3, Linux

- Engineered a **Software-Defined Networking (SDN) controller** on a **k=4 Fat-Tree data center topology**, implementing **real-time link-state monitoring** via **OpenFlow PortStats polling** and **graph-based path computation** (NetworkX k-shortest-paths) for dynamic **traffic engineering**.
- Developed a **bandwidth-aware dynamic routing algorithm** computing per-path **bottleneck bandwidth** across precomputed candidate routes, selecting maximum-capacity paths to mitigate **network congestion** and hotspot formation.
- Benchmarked against **Shortest Path, Round Robin, and Weighted Round Robin** algorithms under simulated congestion, evaluating **throughput, link utilization, packet loss, and jitter** across a reproducible test harness.
- Delivered **45% improvement in average link utilization** over static Shortest Path routing (16% vs. 11%) at a measured 17% peak-throughput tradeoff, validating superior **network-wide resource distribution** under load.

AI-Powered Adaptive Authentication & IAM Gateway

Tech: Node.js, Express, MongoDB Atlas, React, JWT, Passport.js

- Built a centralized **SSO/MFA gateway** serving **5,000+ students** across institute-wide applications, supporting

per-app configurable auth methods (**local, OAuth2, SAML2, OIDC**) with **MongoDB TTL-backed session management**.
- Designed a **risk-scoring model** (logistic regression / gradient boosting) using login behavior, geolocation, and **VPN/proxy detection** to dynamically bypass MFA for low-anomaly sessions and enforce step-up verification for high-anomaly ones.
- Engineered auth security infrastructure with **JWT access/refresh tokens**, **bcrypt/HMAC-hashed credential storage**, scheduled key rotation with backward validation, and **CIDR-based IP whitelisting**.
- Implemented **immutable audit logging** across **13+ security event types** to support traceability and post-incident review.

Multi-Client Quiz System with LLM Integration

Tech: C++, TCP Sockets, Multithreading, Concurrency Control, Custom Application-Layer Protocol

- Architected a **scalable multithreaded TCP server** handling **10+ concurrent clients** via a **custom application-layer protocol** with structured **message framing** for real-time delivery, evaluation, and **leaderboard sync**.
- Implemented **server-side caching** with genre-based lookup to eliminate redundant LLM calls, reducing **latency** and boosting **throughput** under concurrent load.
- Built **fault-tolerant connection management** using custom **heartbeat (KEEP-ALIVE/ALIVE_OK) protocol**, enabling **dead-client detection**, **graceful disconnection**, and **session integrity**.

SKILLS

Programming Languages: C, C++, Python

Web Technologies: HTML, CSS, JavaScript, React, Node.js, Express.js, MongoDB, MySQL

Libraries/Tools: NumPy, Matplotlib, SciPy, Jupyter Notebook, Streamlit , Scikit-learn

EXTRACURRICULAR

- **Vice Captain of the IIT Hyderabad Hockey Team** - Represented the institute in the prestigious Inter I**IT Sports Meet 2024 and 2025** , contributing to **team leadership** and performance at the national collegiate level.

- Led a team of **300+ hostel members** to victory in **IIT Hyderabad's General Championship 2024**, securing **1st place** out of **17 hostels** across **54 cultural, technical, and sports competitions**.